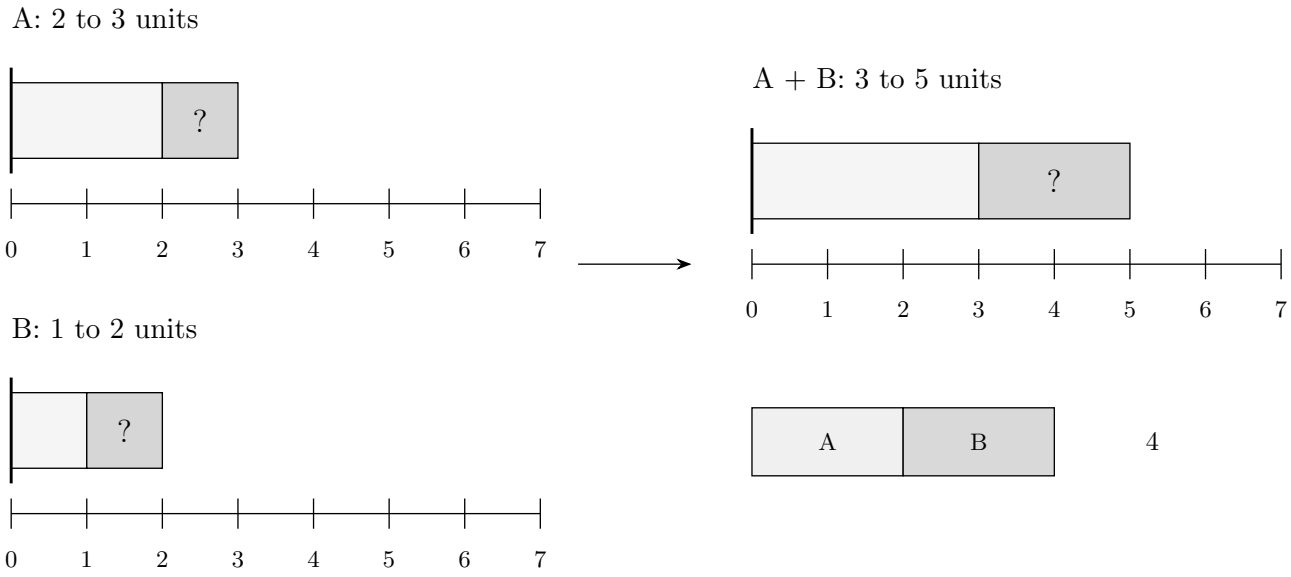
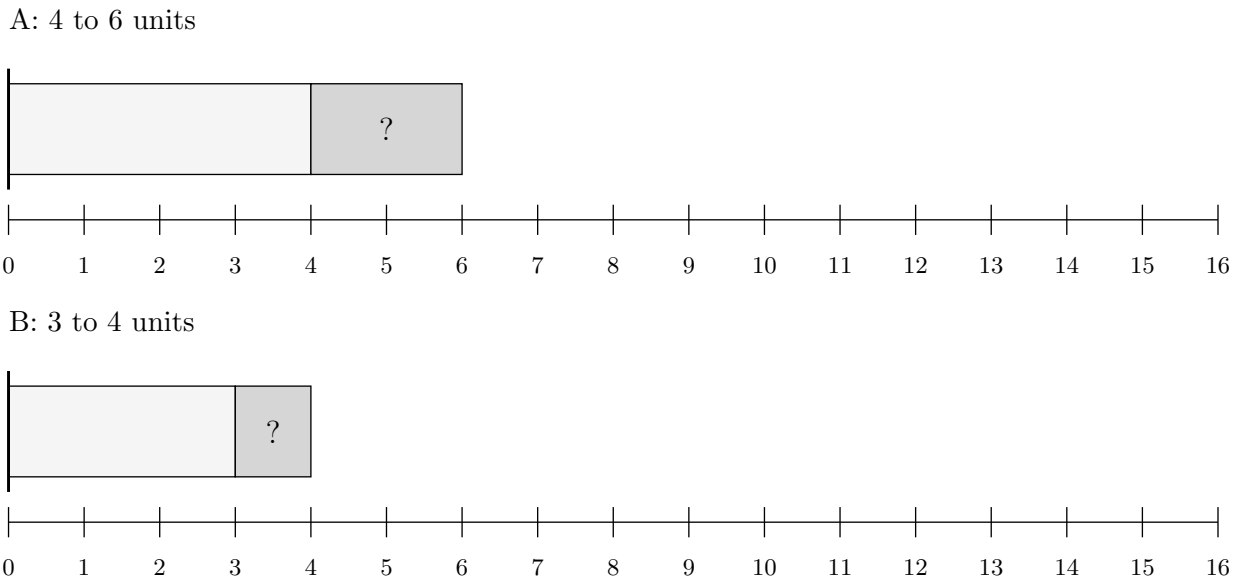
A hidden end can be anywhere in its gray band, including the edges. Each trial may use new lengths. Join strips end to end from 0, with no gap or overlap.

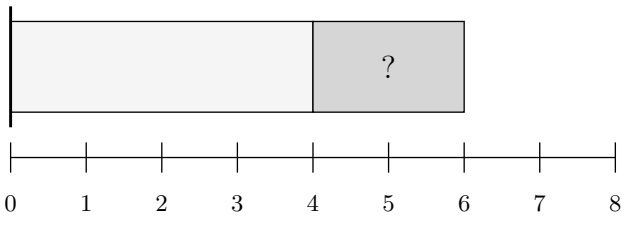


Problem 1: Find the tightest guaranteed range for the join. Someone claims it must end between 8 and 9; build allowed strips that defeat that claim.

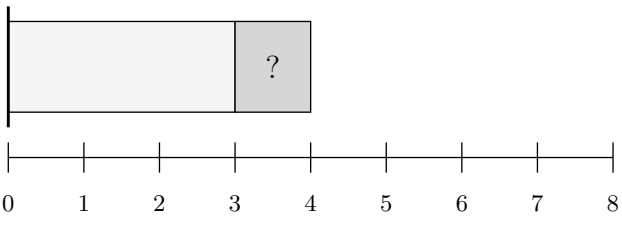


Problem 2: Try only A with B, C with D, and E with F. Choose a pair whose join always reaches 8 and fits within 10. For each rejected pair, let a partner defeat one promise with allowed lengths.

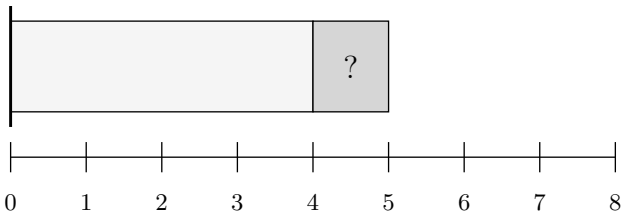
A: 4 to 6 units



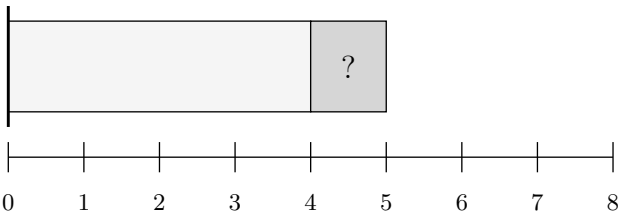
B: 3 to 4 units



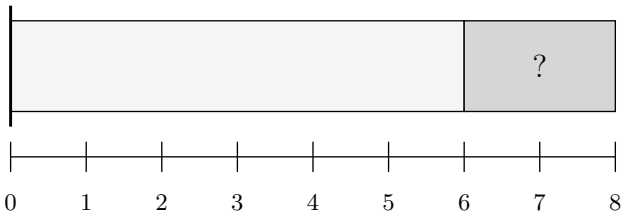
C: 4 to 5 units



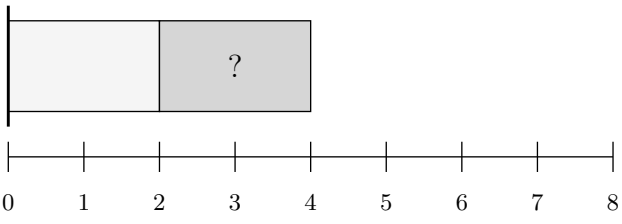
D: 4 to 5 units



E: 6 to 8 units



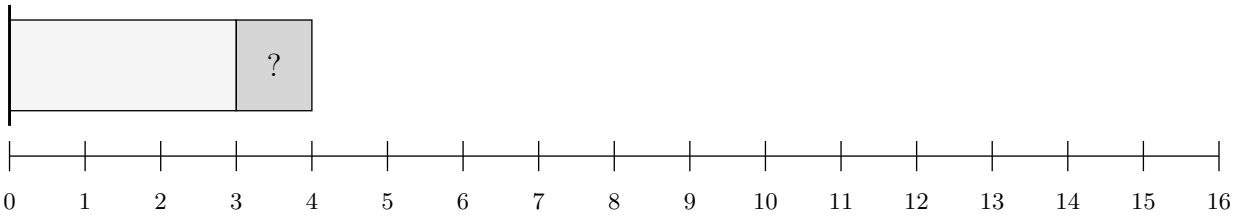
F: 2 to 4 units



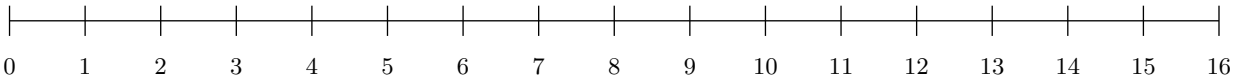
Reaching 8 includes ending at 8. Fitting within 10 includes ending at 10.

Problem 3: A is 3 to 4 units long. Make a range card for B so every join reaches 7 and fits within 9. Make B's allowed range as wide as possible, and explain why it cannot be wider.

A: 3 to 4 units



B:

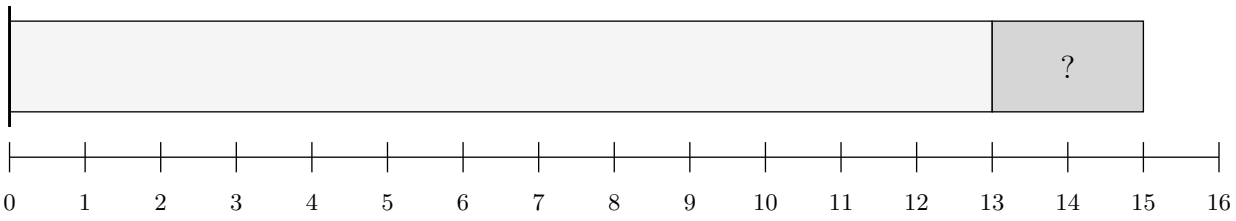




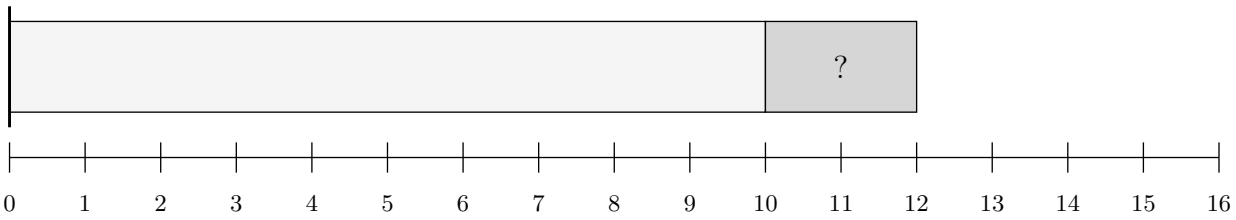
Problem 4: A different pair has A from 4 to 6 and B from 3 to 4. You may learn one exact length. Which strip would you uncover to leave the narrowest guaranteed range for the join?

Problem 5: Align both starts at 0. Find the tightest guaranteed range for the gap between the ends. Build lengths that show neither bound can be improved.

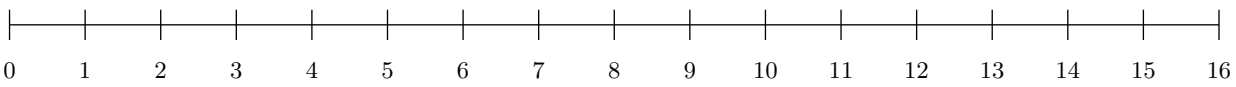
long strip: 13 to 15 units



short strip: 10 to 12 units

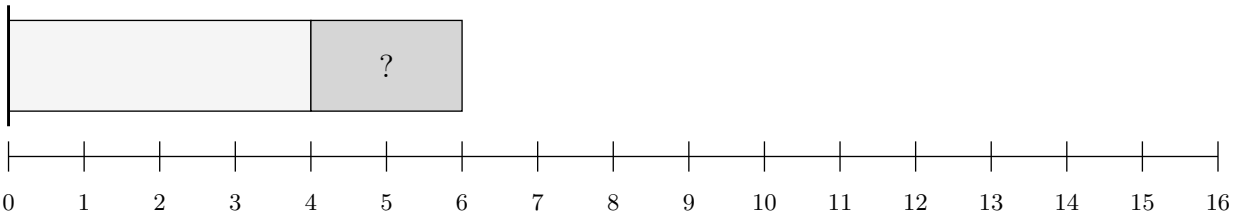


Problem 6: Someone subtracts the two middle values and announces one exact gap. Can you choose allowed strips that show why this is unsafe?

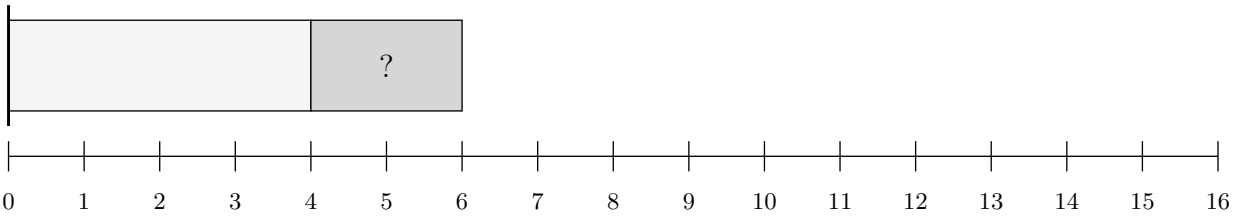


Problem 7: Compare the gaps between the right ends of these joins, starting at 0. Repeated A means the same unchanged length; B may vary separately. Find the tightest range for each gap.

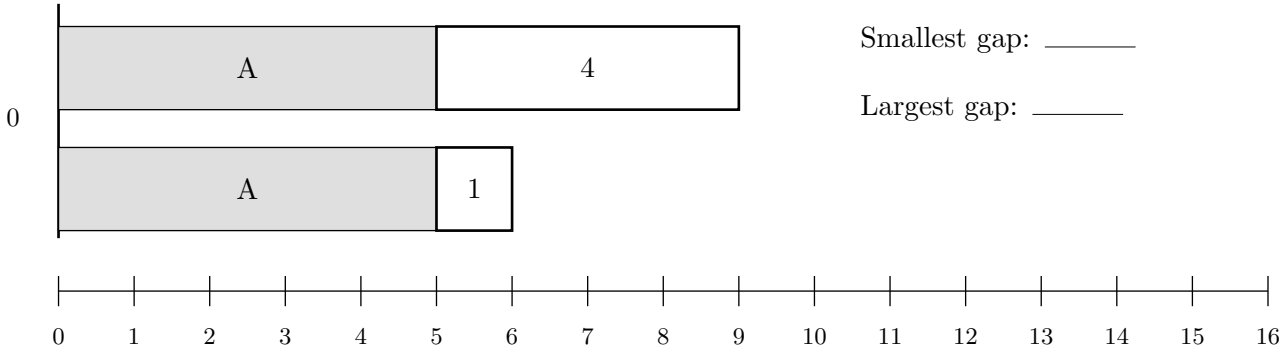
A: 4 to 6 units



B: 4 to 6 units

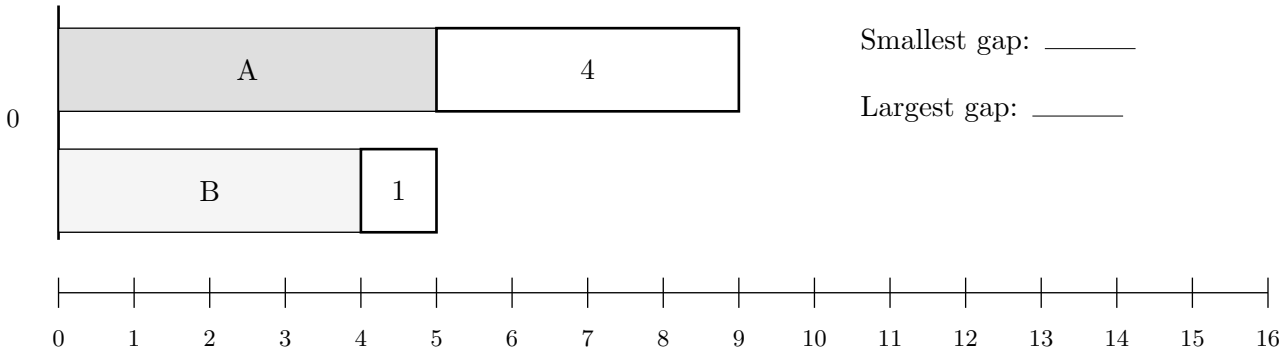


The white pieces are exactly 4 and 1 units long.



Smallest gap: _____

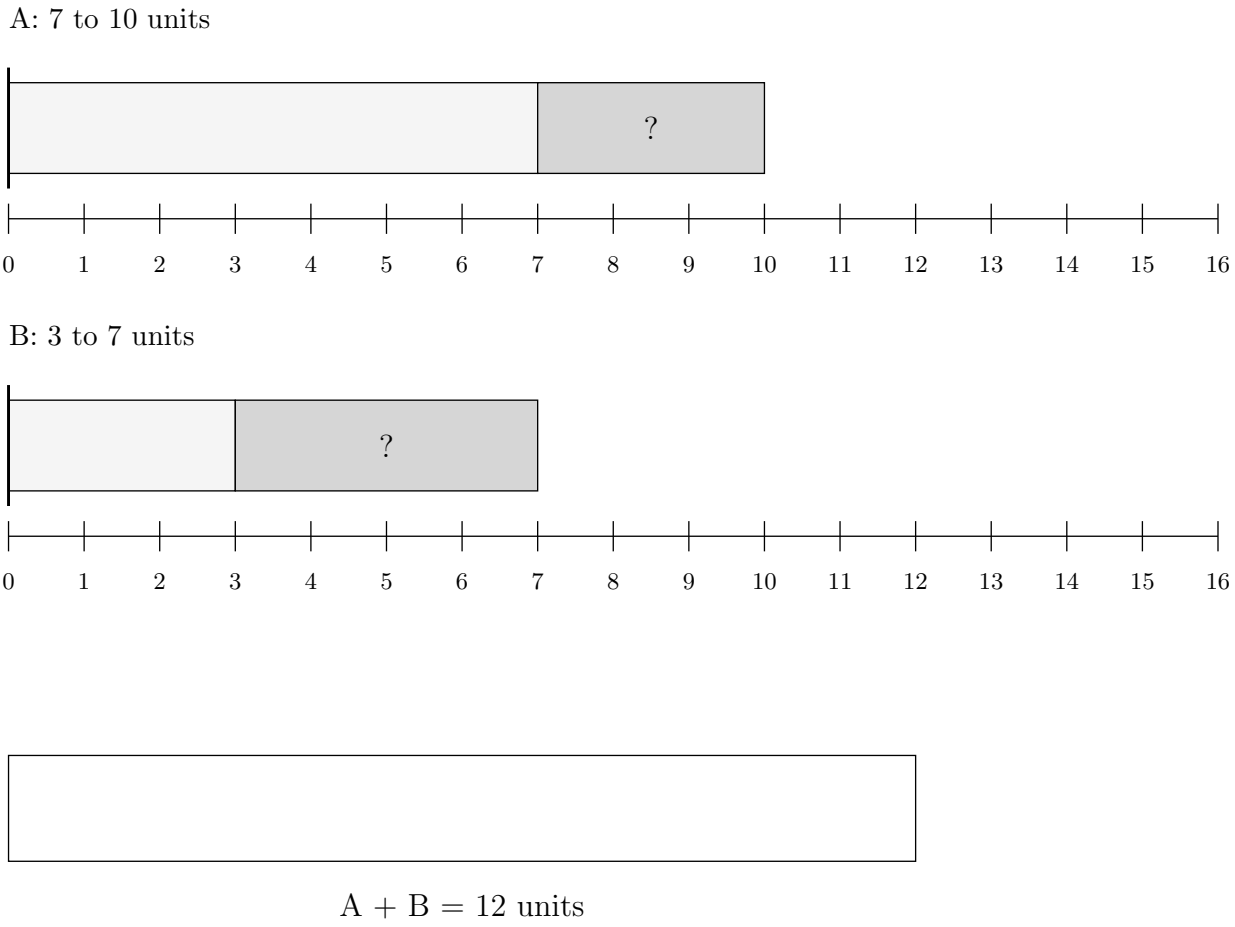
Largest gap: _____



Smallest gap: _____

Largest gap: _____

Problem 8: A is 7 to 10 units and B is 3 to 7 units. You learn that their join is exactly 12 units. Align their starts at 0. Find the smallest and largest possible gap between their ends.



Problem 9: Join A (4 to 6 units) to B (3 to 4 units), then remove that very same A. Someone uses the join's range and A's range separately to claim the remainder could be 1 to 6 units. Is that the tightest range? Explain with the pieces.